

Chapter 2

Shopsphere – Online E-Commerce Database Management System

2.0 Introduction

The ShopSphere Online E-Commerce Database Management System is developed to manage and organize the data required for an online shopping platform. The system maintains information related to customers, sellers, products, orders, payments, deliveries, inventory, and feedback. Each entity is identified along with its attributes, primary keys, and foreign keys to ensure proper data organization and efficient database management. The relationships between the entities are analyzed to represent the interactions within the system and to establish a well-structured relational database design. This analysis helps in developing the Entity Relationship (ER) Diagram and provides a strong foundation for database implementation while ensuring data integrity, consistency, and minimal data redundancy.

2.1 Entities and Attributes

Entity: Customer

Attribute	Key Type
Customer_ID	Primary Key
Full_Name	Not Null
Email	Unique Key
Mobile_Number	Unique Key
Password	Not Null
Registration_Date	-

Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null

Product_Price	-
Product_Description	-
Stock_Availability	-
Seller_ID	Foreign Key

Entity: Seller

Attribute	Key Type
Seller_ID	Primary Key
Store_Name	Not Null
Seller_Email	Unique Key
Contact_Number	-
Business_Type	-

Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Order_Status	-
Delivery_Type	-

Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key

Payment_Mode	-
Payment_Status	-
Transaction_Date	-

Entity: Delivery

Attribute	Key Type
Delivery_ID	Primary Key
Order_ID	Foreign Key
Delivery_Address	-
Delivery_Status	-
Expected_Date	-

Entity: Inventory

Attribute	Key Type
Inventory_ID	Primary Key
Product_ID	Foreign Key
Available_Quantity	-
Reorder_Level	-

Entity: Feedback

Attribute	Key Type
Feedback_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Check
Comments	-

2.2 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality
Customer	Places	Order	One-to-Many
Seller	Lists	Product	One-to-Many
Order	Contains	Product	Many-to-Many
Order	Has	Payment	One-to-One
Order	Has	Delivery	One-to-One
Product	Stored In	Inventory	One-to-One
Customer	Gives	Feedback	One-to-Many
Product	Receives	Feedback	One-to-Many

2.3 Primary Keys and Foreign Keys

Primary Keys

- Customer_ID
- Product_ID
- Seller_ID
- Order_ID
- Payment_ID
- Delivery_ID
- Inventory_ID
- Feedback_ID

Foreign Keys

- Seller_ID in Product
- Customer_ID in Order
- Order_ID in Payment

- Order_ID in Delivery
- Product_ID in Inventory
- Customer_ID in Feedback
- Product_ID in Feedback

2.4 Conclusion

The entities, attributes, keys, and relationships of the ShopSphere Online E-Commerce Database Management System have been identified and analyzed. This design provides the foundation for creating the ER Diagram and implementing the database in the next phase of the project.